COMPLETE SCIENTIFIC WORKFLOW & MATHEMATICAL FRAMEWORK

Controlled Disentanglement of Synthetic Edge Seams vs. Macro-Spatial Context in Deep Brain MRI Classifiers

STAGE 1: DATASET & SPLITS

• Source: Cheng Figshare (1512427)
• Samples: 3,064 2D T1-CE MRI slices
• Patient Clusters: 233 unique IDs
• Classes:
  - Meningioma: 708 slices (23.1%)
  - Glioma: 1,426 slices (46.5%)
  - Pituitary: 930 slices (30.4%)

Cross-Validation Scenarios:
• S0: Patient-Disjoint 5-Fold
  - Grouped strictly by Patient ID
• S1: Candidate Cleaned (SSIM ≥ 0.80)
  - 231 merged group clusters
• S2: Aggressive Grouping (SSIM ≥ 0.75)
  - 217 merged group clusters

Formula Used:
Structural Similarity Index (SSIM)
for near-duplicate audit

STAGE 2: MODEL TRAINING & FREEZING

• Architecture: ResNet-50 (Pretrained)
• Total Checkpoints: 15 Frozen Checkpoints
  (3 Scenarios × 5 Folds)
• Input Dimensions: 512×512 → 224×224
• Preprocessing:
  - Min-Max normalization to [0, 1]
  - ImageNet mean & std standardisation
• Loss Function: Cross-Entropy Loss
• Optimizer: AdamW, Cosine LR Decay
• Early Stopping on Validation Fold

ZERO-LEAKAGE INVARIANT:
Weights frozen after Phase 2.
Zero fine-tuning or retraining during
subsequent spatial interventions!

STAGE 3: CONTROLLED INTERVENTIONS

PROTECTED ZONE INVARIANT:
Tumour Core + 15px Margin = 100% UNTOUCHED
M_protect = Dilate(Tumour_Mask, r=15px)

4 Matched Conditions:
1. C0: Unmodified Baseline
  - Natural clinical MRI slice

2. C1: Seam-Only Artifact Control
  - Synthetic boundary steps on unpermuted tiles
  - Matches scrambled DC offset jump

3. C2: Seam-Suppressed Rearrangement
  - Permuted 16×16 distant tiles
  - 4px Gaussian cross-fading on boundaries
  - Seam Sobel -54.8%, Fourier HF -84.9%

4. C3: Raw Historical Scramble (C5LP)
  - Distant tiles permuted without smoothing
  - Both seams + spatial layout disrupted

STAGE 4: PREDICTION ACCOUNTING

Out-of-Fold (OOF) Forward Inference:
• 3 Scenarios (S0, S1, S2)
• 4 Conditions (C0, C1, C2, C3)
• 3,064 Held-Out Test Slices
• Total Forward Evals: 36,768 records

Evaluation Unit:
• Strictly out-of-fold slice-level predictions
• Zero slice reuse across train/test
• Zero test-time ensembling across folds

Primary Metric: Macro-Averaged F1
Macro-F1 = (F1_men + F1_gli + F1_pit) / 3

Performance Outcomes (S0):
• C0 Baseline: 0.9333
• C1 Seam-Only: 0.6290 (Drop: +0.3043)
• C2 Seam-Supp: 0.4525 (Drop: +0.4808)
• C3 Raw Scram: 0.5118 (Drop: +0.4215)

COMPLETE MATHEMATICAL & STATISTICAL FORMULAS APPLIED IN THE RESEARCH

1. PROTECTED MARGIN DILATION
M_protect(x,y) = M_core ⊕ B_r (where r = 15 pixels)
• Purpose: Guarantees 100% preservation of tumour and peritumoral tissue.
• Applied in: Phase 3, Phase 5.1, Phase 9 (geometry generator).

2. SEAM-SUPPRESSED CROSS-FADING (C2)
I_C2(x,y) = α(x,y) · T_k(x,y) + (1 - α(x,y)) · T_m(x,y)
where α(x,y) = 0.5 · [1 + erf((d(x,y)) / (σ√2))], σ = 4.0 px
• Purpose: Suppresses boundary step discontinuities during tile rearrangement.
• Applied in: Phase 9 intervention synthesis (removes high-frequency artifact confound).

3. SPATIAL SOBEL BOUNDARY GRADIENT ENERGY
E_Sobel = (1 / |Ω_seam|) ∑_{(x,y) ∈ Ω_seam} √((∂I/∂x)^2 + (∂I/∂y)^2)
• Measured: C0 = 31.48 | C1 = 120.58 | C2 = 76.00 (-54.8%) | C3 = 129.90
• Applied in: Image diagnostics, Phase 9.1 audit & Table 7.

4. HIGH-FREQUENCY FOURIER RADIAL SPECTRAL ENERGY
E_HF = ∫_{r > r_c} ∫_0^{2π} |F(r cos θ, r sin θ)|^2 r dθ dr, r_c = 0.5 · (N/2)
• Measured: C0 = 421.3 | C1 = 2,342.1 | C2 = 715.4 (-84.9%) | C3 = 2,597.4
• Applied in: Frequency domain artifact verification (Table 7 & Figure 8).

5. CANONICAL PAIRED CONTRAST (A-minus-B Convention)
Δ = Macro-F1(A) - Macro-F1(B)
• Rule: Positive Δ indicates performance drop in condition B vs baseline A.
• Applied in: Table 6, Figure 6, hypothesis testing.

6. SYNCHRONIZED CLUSTER BOOTSTRAP (B = 2,000)
For b = 1 ... B: Resample Patient Clusters C_1*, ..., C_K* with replacement.
Evaluate matched out-of-fold slices:
Δ_b* = Macro-F1_b*(A) - Macro-F1_b*(B)
95% CI = [Percentile_{2.5%}(Δ*), Percentile_{97.5%}(Δ*)]
• Applied in: Estimating exact uncertainty while preserving patient correlation.

7. MCNEMAR PAIRED DISAGREEMENT TEST (Continuity Corrected)
χ^2 = (|b - c| - 1)^2 / (b + c)
where b = Correct(A) ∧ Incorrect(B), c = Incorrect(A) ∧ Correct(B)
• Purpose: Hypothesis test of sample-level prediction shifts.
• Applied in: Primary hypothesis validation (C3 vs C1: χ^2 = 170.5, p < 10^-15).

8. HOLM-BONFERRONI MULTIPLICITY CONTROL (FWER α = 0.05)
Sort p-values: p_(1) ≤ p_(2) ≤ ... ≤ p_(m) (m = 15 hypotheses).
Reject H_(k) if p_(k) ≤ α / (m - k + 1).
• Applied in: Simultaneous control across all 15 confirmatory contrasts.

9. DESCRIPTIVE MARGINAL PERFORMANCE DROP RATIO
Ratio = Δ_{C1} / Δ_{C3} = (+0.3043) / (+0.4215) = 72.2%
• Epistemic Guardrail: Descriptive ratio only; NOT an additive variance split!